

Enabling precision grasping and tactile feedback in tetraplegia using a bidirectional brain-computer interface and a low-profile active orthosis

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INTRODUCTION

- Individuals living with spinal cord injury (SCI) prioritize restoring upper limb function¹
- Implanted microelectrode arrays in motor areas can provide access to a user's motor intent for a brain-computer interface (BCI)
- Intra-cortical microstimulation (ICMS) can be used as tactile feedback in a sensorimotor BCI
- Transcutaneous spinal cord stimulation (tSCS) can be used to improve muscle control in SCI patients

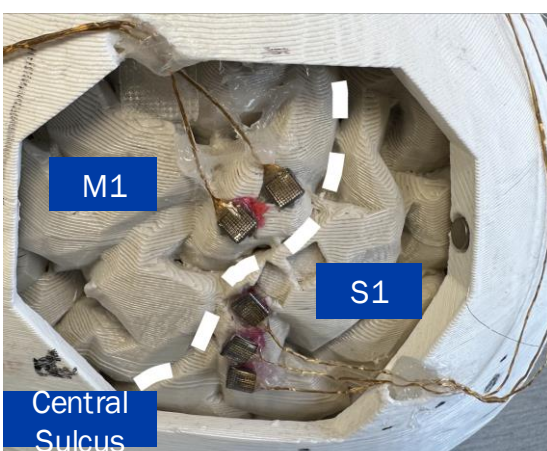
A BCI-mediated active orthosis, paired with tSCS motor improvements, could be a novel approach to enable dexterous control of the fingers, grasp, and reach

Objective: Restore sensorimotor grasp in a SCI participant using a BCI-mediated active orthosis, inspired by the spinal cord, for dexterous motor control with somatosensory feedback.

METHODS

Participant: 47-year-old male presenting with a C5 SCI.

- 5 Utah Arrays, implanted in 2023
- tSCS was delivered at C5 (~1h) and C7 (~30m) roots at 300mA for 10 weeks.
- Scans of the participant's right forearm and fingers were taken to 3D print a custom orthosis (PLA) and finger rings (resin).

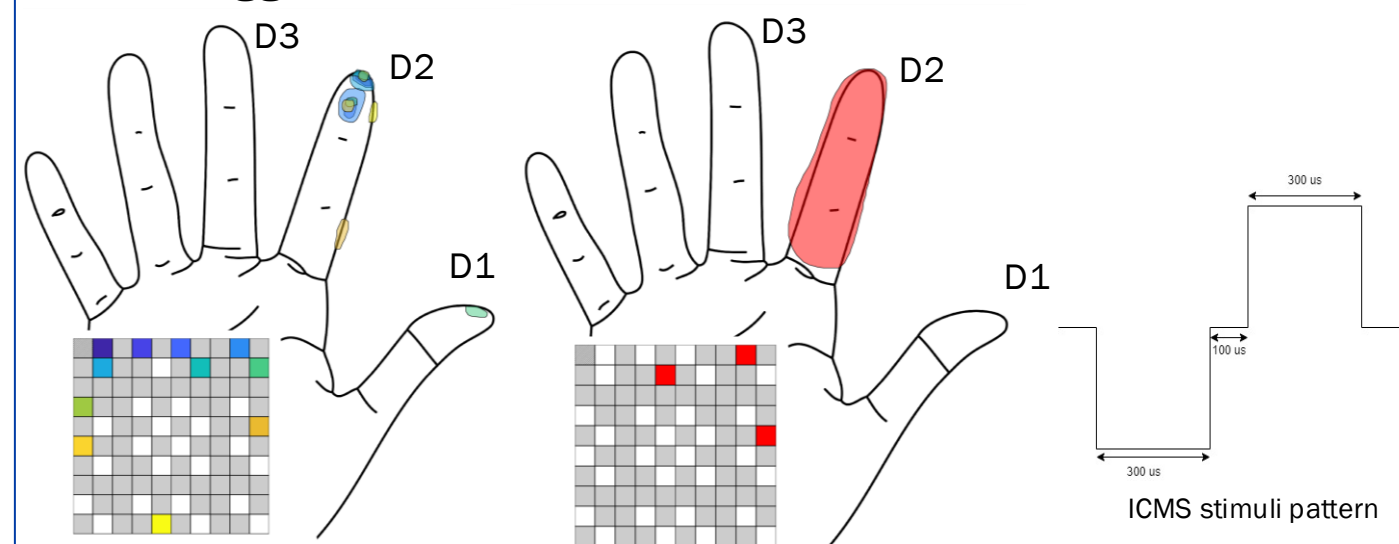


Motor Approach

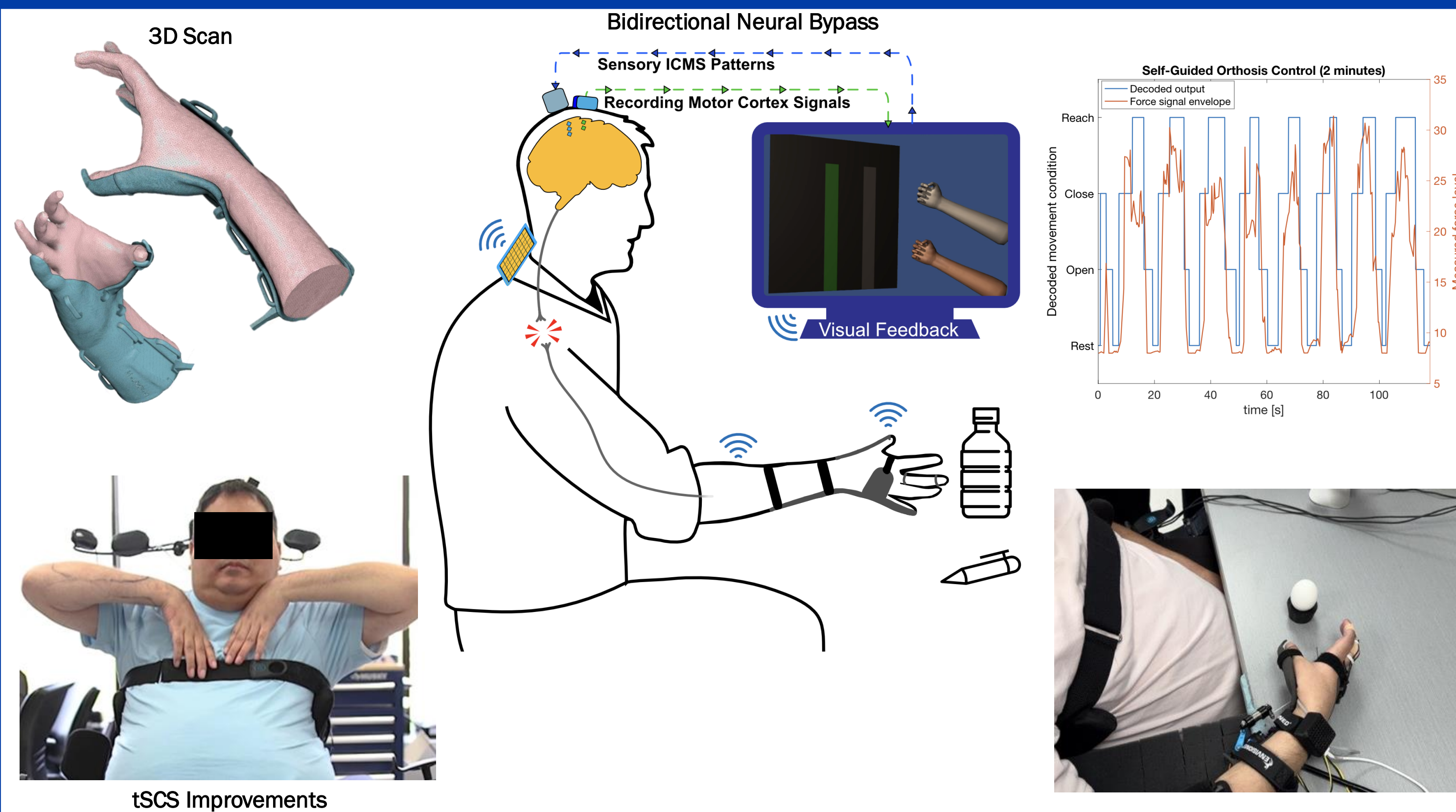
- An LSTM regression decoder was used to predict motor intention and grasp intensity.
- Decoded outputs were sent to a servo motor to execute the desired grasp intensity.

Sensory Approach

- An embedded force sensor was used to record thumb pressure against a target object.
- Force readings above a defined threshold would halt decoding and trigger ICMS in S1.



MECHANISM



CONCLUSION & FUTURE WORK

- An active orthosis can successfully facilitate fine finger movement in tetraplegia
- Orthosis is adaptable and was used to grasp fragile objects

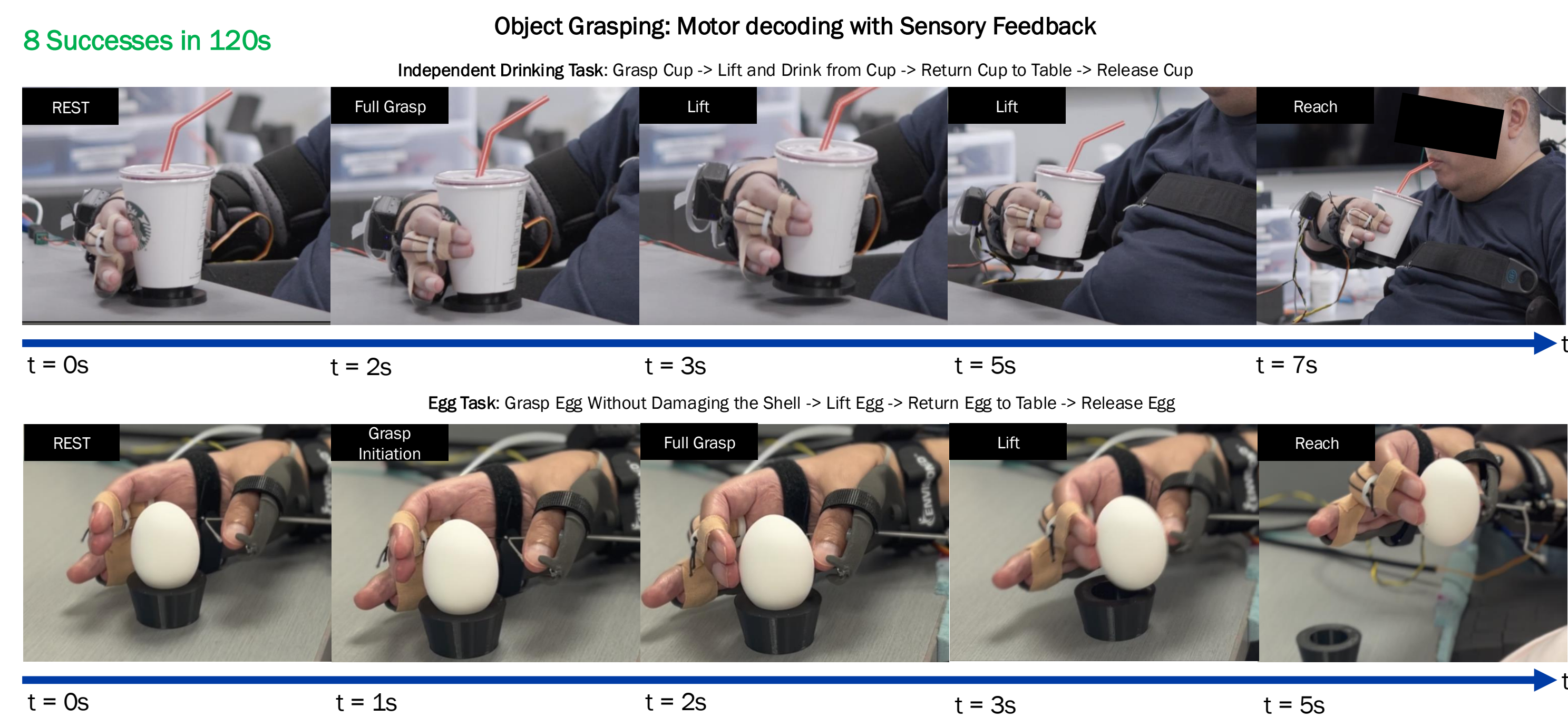
Limitations:

- D3 best for motor but D2 best for sensory, requiring fingers to be tethered together.
- A tactile sensor was placed on D1 which ensured constant force output upon object contact but perception was on D2.

Future Work:

- Execute grasp and lift, without visual cues, using only ICMS as sensory feedback
- Refine decoder for faster closed-loop behavior and reaction.
- Enable dynamic wrist movements
- Enable dynamic grasping (2-step, e.g., toothpaste tube)
- Incorporate 3 levels of force into grasping BCI
- Create a more general orthosis fitting
- More therapeutic tSCS to strengthen upper limb control

RESULTS



REFERENCES

- [1] Anderson, K. D. (2004). Targeting recovery: priorities of the spinal cord-injured population. *J. Neurotrauma* 21, 1371–1383. doi: 10.1089/neu.2004.21.1371

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